Credit Risk Default Prediction: Data Processing Pipeline

CS611

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Outline

Data

- Exploratory Analysis
- Pre-Processing

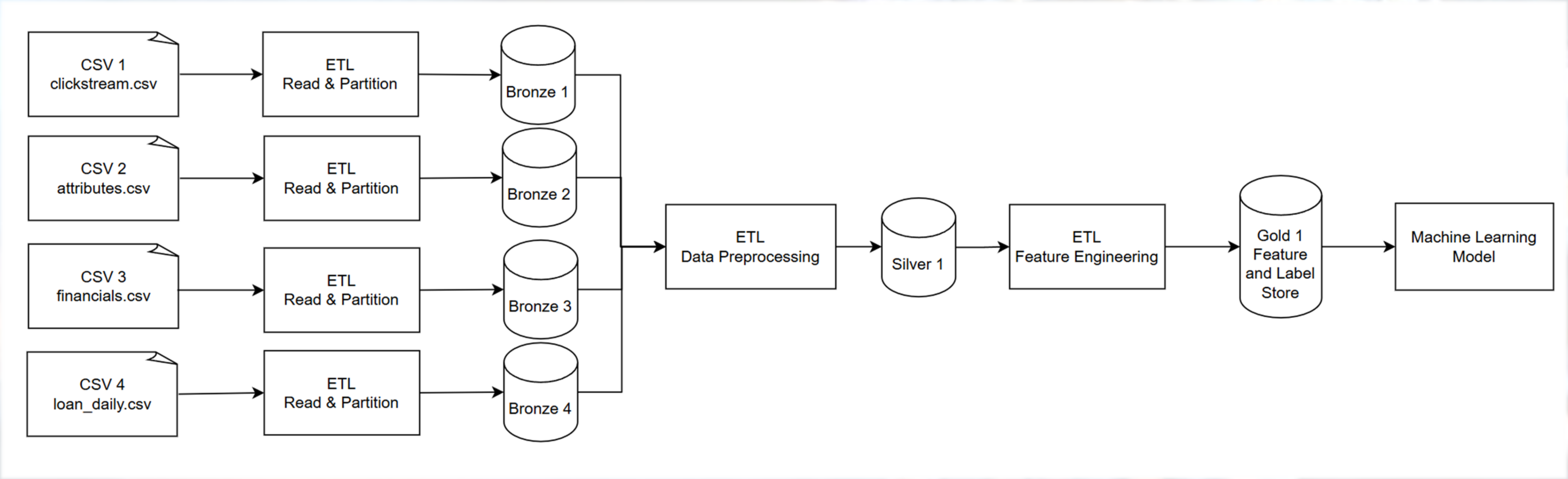
Medallion Architecture

- Data Pipeline
- Feature Store

Machine Learning

- Model Choice
- Outcome

Data Pipeline



Exploratory Data Analysis

Data Issues

- 756 customer IDs with formatting errors, i.e. broken IDs
  - Affecting more than 8,000 data rows (LMS_Loan_Daily)
- Data points with invalid characters ("_", symbols such as "#@!")
- Impossible / unlikely outliers
  - 8,678 years old for Age
  - Negative number of bank accounts
  - $23.8m in annual income (unlikely to take small-value loans)
  - Poor loan description, e.g. 9 different loan types concatenated together

Bronze Layer

Design: As-Is Data Ingestion with Monthly Partition

- Four CSVs were ingested into four separate Bronze tables without any modification in parquet format for faster read/write speeds
- Maintains immutable records as back-up against downstream failure or error
- Monthly partition to facilitate monthly batch-load in production

Silver Layer

Design: Data Pre-Processing for Robust Machine Learning

- Cleaned customer_ID
  - Fixed to standardised length and corrected formatting (no whitespaces etc.)
  - Checked for unique IDs and ensure no rows with empty values after joining downstream
- Checked for outliers and dirty data
  - Stripped non-numeric text for numerical features such as Age, removed outliers above 100
  - Capped number of accounts to 0 and 50
  - Cleaned categorical data by removing “NM “ values etc. into “No Loan”, converted string for multiple loans to new variable to flag customers with multiple loans
  - Data entry errors such as “_” changed to “unknown”
- Calculate Days Past Due (current snapshot date – first payment missed date)
- Outer Join to combine each customer’s attributes to every active instalment row

Gold Layer

Design: Feature Engineering and Check for Data Leakage

- Preventing Data Leakage
  - Dropped features which use data not available at time of prediction, i.e. dpd, overdue_amt etc.
- Transitioned from disparate datasets into a fixed-horizon matrix which facilitates binary classification ML model training
- Target Engineering: label column calculated using dpd >= 90 days (1 = default, 0 for no default)
- Feature Optimisation: dropped features such as Name and SSN which are not useful for ML modelling
  - Nulls replaced with 0 to safeguard Random Forest algorithm
- Output naming convention standardisation with serialisation for clean handoff to ML pipeline

Machine Learning Model – Random Forest

Objective: Diagnostic Validation for Data Pipeline Architecture

- Simple binary classification algorithm
  - Ingested the Gold feature store
  - Target: Gold label store
- Pipeline Validation
  - Gold layer successfully loaded into PySpark environment
- Data architecture stable and ready for further ML model testing